Patient and mutation filtering analysis for actionable mutation modeling

Overview

This document presents data to inform decisions on filtering and aggregating clinically actionable mutations. The data are primarily aimed at aggregations at the level of individual patients. The results are organized into three sections

1. Part 1 shows results for LEVEL 2 actionable events as assigned by OncoKB. The most common mutations are described at the level of cancer type, gene, and amino acid change.
2. Part 2 shows results related to mutation co-occurrence at the level of Level 1 and Level 2 events
3. Part 3 shows data on read coverage data and how filtering on this metric could affect patient-level aggregation.

Results are restricted to a subset of cancer types that have 250 or more patients with actionable events.

Part 1: Count of OncoKB Level 2 actionable events

Most common Level 2 events by cancer type

Cancer subtypes included in ML modeling ordered by most frequent level 2 events.

cancer type detailed	count and perc of actionable events
Breast Invasive Lobular Carcinoma	88 (8.7%)
Cutaneous Melanoma	46 (6.4%)
Bladder Urothelial Carcinoma	48 (6.2%)
Melanoma	96 (6.2%)
Breast Mixed Ductal and Lobular Carcinoma	19 (6.1%)
Colon Adenocarcinoma	69 (5.4%)
Colorectal Adenocarcinoma	14 (4.9%)
Invasive Breast Carcinoma	78 (4%)
Breast Invasive Cancer, NOS	10 (3.5%)
Breast Invasive Ductal Carcinoma	144 (3.2%)
Intrahepatic Cholangiocarcinoma	8 (2.8%)
Breast Invasive Carcinoma, NOS	8 (2.8%)
Non-Small Cell Lung Cancer	18 (2.6%)
Gastrointestinal Stromal Tumor	37 (2.4%)
Lung Adenocarcinoma	89 (0.9%)
Papillary Thyroid Cancer	NA (NA%)
Prostate Adenocarcinoma	NA (NA%)
Acute myeloid leukemia	NA (NA%)
Oligodendroglioma	NA (NA%)
High-Grade Serous Ovarian Cancer	NA (NA%)
Astrocytoma	NA (NA%)

Most common Level 2 events for each cancer type & gene

Top 15 most common Level 2 events by gene and cancer type:

cancer type detailed	gene	count and perc of actionable events
Breast Invasive Lobular Carcinoma	ERBB2	86 (8.5%)
Cutaneous Melanoma	KIT	46 (6.4%)
Bladder Urothelial Carcinoma	FGFR3	48 (6.2%)
Melanoma	KIT	96 (6.2%)
Colon Adenocarcinoma	POLE	66 (5.2%)
Breast Mixed Ductal and Lobular Carcinoma	ERBB2	16 (5.1%)
Colorectal Adenocarcinoma	POLE	14 (4.9%)
Invasive Breast Carcinoma	ERBB2	71 (3.6%)
Intrahepatic Cholangiocarcinoma	KRAS	8 (2.8%)
Gastrointestinal Stromal Tumor	PDGFRA	37 (2.4%)
Breast Invasive Ductal Carcinoma	ERBB2	104 (2.3%)
Breast Invasive Cancer, NOS	ERBB2	6 (2.1%)
Breast Invasive Carcinoma, NOS	ERBB2	5 (1.7%)
Breast Invasive Cancer, NOS	AKT1	3 (1%)
Breast Invasive Carcinoma, NOS	AKT1	3 (1%)

Most common Level 2 events for each cancer type, gene, and HGVSp entry

Top 15 most common Level 2 events by gene, cancer type and amino acid change.

cancer type detailed	gene	protein mutation	count and perc of actionable events
Melanoma	KIT	p.L576P	40 (2.6%)
Breast Invasive Lobular Carcinoma	ERBB2	p.L755S	32 (3.1%)
Colon Adenocarcinoma	POLE	p.P286R	26 (2%)
Invasive Breast Carcinoma	ERBB2	p.L755S	23 (1.2%)
Lung Adenocarcinoma	ALK	p.G1202R	21 (0.2%)
Breast Invasive Ductal Carcinoma	ERBB2	p.G778_P780dup	20 (0.4%)
Lung Adenocarcinoma	ALK	p.L1196M	19 (0.2%)
Gastrointestinal Stromal Tumor	PDGFRA	p.V561D	19 (1.3%)
Cutaneous Melanoma	KIT	p.L576P	18 (2.5%)
Breast Invasive Ductal Carcinoma	ERBB2	p.L755S	17 (0.4%)
Breast Invasive Ductal Carcinoma	ERBB2	p.V777L	17 (0.4%)
Colon Adenocarcinoma	POLE	p.V411L	16 (1.3%)
Invasive Breast Carcinoma	ERBB2	p.V777L	15 (0.8%)
Melanoma	KIT	p.K642E	15 (1%)
Cutaneous Melanoma	KIT	p.K642E	14 (2%)

Part 2: Actionable mutation co-occurrence across OncoKB levles

To investigate the rate of co-occurrence events in AACR GENIE, we tabulate the number of patients across all cancer types that had actionable events at different OncoKB Levels. Any entry with a value >1 or a row with more than one positive value represents a actionable mutation co-occurrence found in a given patient.

Level 1 Event Count	Level 2 Event Count	Any Level Event Count	Number of patients
0	0	0	175777 (82.8%)
1	0	0	31743 (15%)
2	0	0	2256 (1.1%)
0	1	0	1753 (0.8%)
1	1	0	263 (0.1%)
3	0	0	227 (0.1%)
0	2	0	75 (0%)
4	0	0	29 (0%)
2	1	0	28 (0%)
1	2	0	12 (0%)
5	0	0	9 (0%)
6	0	0	4 (0%)
2	2	0	3 (0%)
3	1	0	2 (0%)
0	3	0	1 (0%)

Part 3: Effect of coverage filtering on the rate of actionable mutations

To examine the impact of coverage we, calculate the tumor profiling coverage as the following: t_ref_count+t_alt_count. We tested to see the impact of filtering out variants with a min coverage below 20 bp.

The count of variant events below 20bp of coverage was as follows:

```
  FALSE    TRUE
2688547   16768
```

This resulted in the following impact on actionable mutation count:

```
[1] "Level 1 actionable events before filtering: "
```

```
[1] 34580
```

```
[1] "Level 1 actionable events after filtering: "
```

```
[1] 33323
```

```
[1] "percent difference: "
```

```
[1] "-3.635%"
```

And the following counts change in counts for Level 2 events:

```
[1] "Level 2 actionable events before filtering: "
```

```
[1] 1830
```

```
[1] "Level 2 actionable events after filtering: "
```

```
[1] 1778
```

```
[1] "percent difference: "
```

```
[1] "-2.842%"
```